

Applied Bioinformatics Laboratory

Experiment – 1

A: Retrieve and analyze glycan-related information using Glyco@ExPASy.

Question:

Glyco@ExPASy is an integrated bioinformatics resource for glycomics that provides information on glycoproteins, glycosylation sites, enzymes involved in glycan biosynthesis, and carbohydrate structures. It supports the study of how glycans influence protein function, stability, and biological activity.

AIM:

To collect and examine glycan-associated data using the Glyco@ExPASy bioinformatics platform.

BACKGROUND:

Glycosylation is an essential post-translational modification that significantly influences protein folding, structural stability, cell communication, and disease progression. Since laboratory-based glycosylation studies are technically challenging and time-consuming, bioinformatics resources such as Glyco@ExPASy provide an efficient alternative for analyzing glycoproteins and

glycan-related characteristics. This experiment illustrates how Glyco@ExPASy can be applied to study the functional relevance of protein glycosylation

METHODOLOGY:

1. Launch a web browser and visit the ExPASy bioinformatics portal.
2. Access the Glyco@ExPASy glycomics resource from the platform.
3. Choose the available databases related to glycoproteins.
4. Obtain detailed information on glycoproteins, including N-linked and O-linked glycosylation sites.
5. Study the location and nature of glycosylation modifications present on selected proteins.
6. Explore data related to enzymes participating in glycan biosynthesis, such as glycosyltransferases and glycosidases.
7. Review the associated glycan and carbohydrate structural information.
8. Relate observed glycosylation patterns to protein function, stability, and biological roles.
9. Document and organize the findings in a clear and systematic manner.

RESULT:



Fig 1: Glyco@ExPASy portal overview

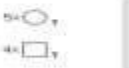
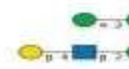
Reported Sites with Glycan			Reported Sites	Predicted Only	Text Mining
Summary: 17 site(s) total, 79 N-linked glycan(s) at 6 site(s), 11 O-linked glycan(s) at 11 site(s) *Not Reported*: No site information available for 19 glycan(s)					
Records per page			20	Showing 1 to 20 of 126 Results	
1 2 3 4 5 > >>					
Published 1 GlyCosmos 1	O-linked	G49632WD		Not Reported	
Published 1 GlyCosmos 1	N-linked	G50045TK		Not Reported	
Published 1 GlyCosmos 1	N-linked	G51367TM		Not Reported	
Published 1	O-	G57321FI		Not	

Fig 2: Reported N-linked and O-linked glycosylation sites in Glyco@ExPASy

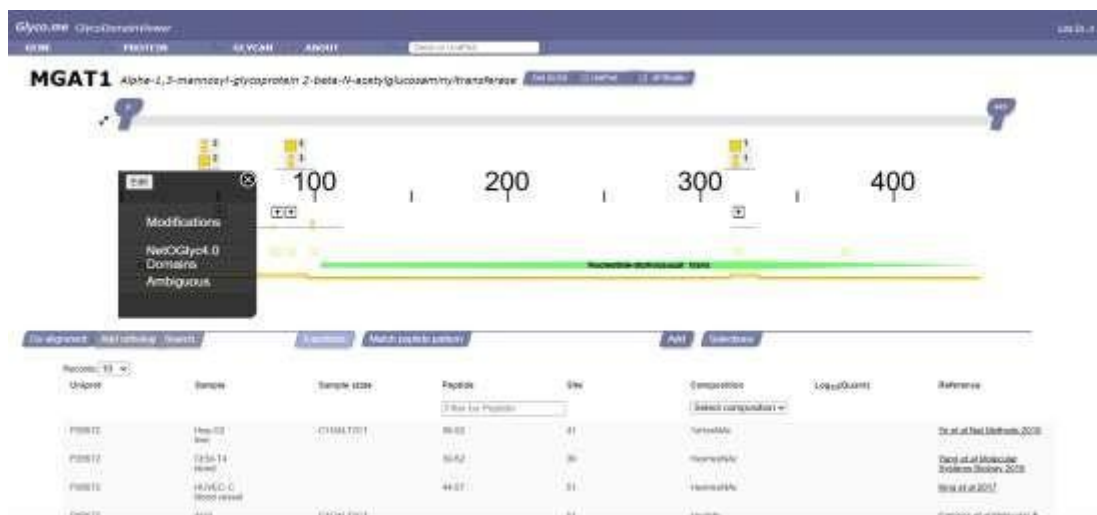


Fig 5: GlycoDomainViewer visualization of MGAT1 glycosylation sites

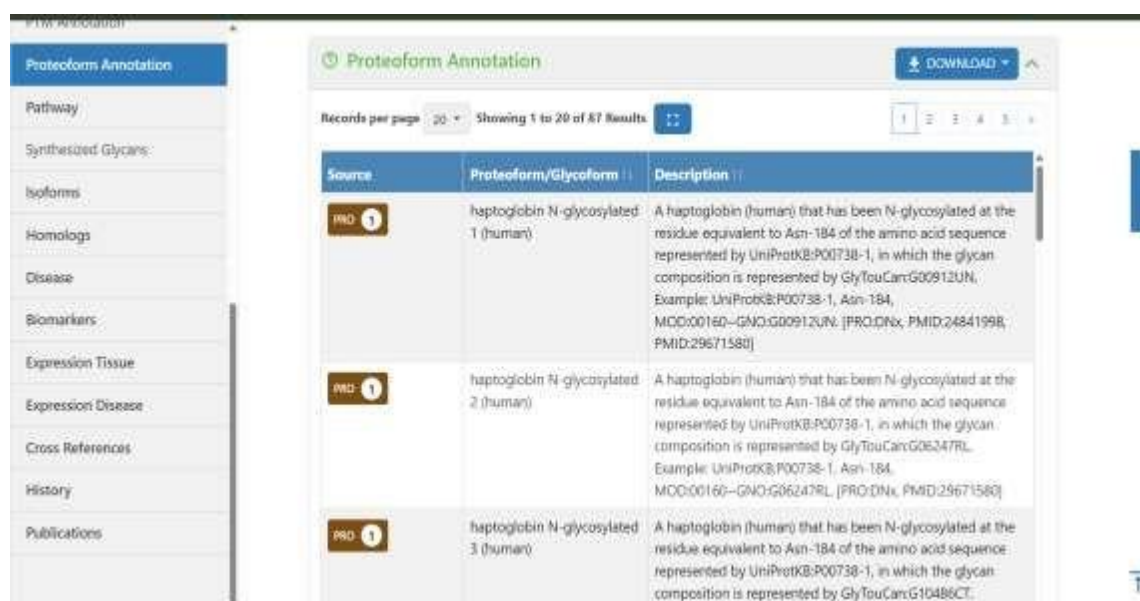


Fig 6: Proteoform annotation results for N-glycosylated haptoglobin

INTERPRETATION:

The analysis performed using Glyco@ExPASy provided detailed insights into protein glycosylation patterns and their functional significance. The platform successfully identified both N-linked and O-linked glycosylation sites on selected glycoproteins, highlighting their specific positions and types of carbohydrate attachments. Information on glycan biosynthesis enzymes, including glycosyltransferases and glycosidases, further explained how these modifications are formed and regulated. The observed glycosylation patterns were found to influence protein folding, stability, and biological activity, demonstrating the critical role of glycans in modulating protein function.